Gamepad 1: DRIVER

Drives the robot. Pick it on the Driver Station with START + A. Same in every TeleOp (DriverControls.java).

Left trigger (LT / L2)
TURBO (hold)
full speed: 100%

Left bumper (LB / L1)
SLOW (hold)
lining up: 40%

Dpad UP
(free)

Dpad LEFT
(free)

Dpad RIGHT
(free)

Dpad DOWN
(free)

Left stick
DRIVE
forward / back, strafe left / right

Left stick: press it down
(free)

Right trigger (RT / R2)
TURBO (hold)
full speed: 100%

Right bumper (RB / R1)
SLOW (hold)
lining up: 40%

Y (triangle)
(free)

X (square)
(free)

B (circle)
(free)

A (cross)
(free)

Right stick
TURN
push left / right

Right stick: press it down
(free)

Three driving speeds

SLOW hold either bumper: 40% power, for lining up
NORMAL nothing held: 70% power, everyday driving
TURBO hold either trigger: 100% power, crossing the field
Both held? SLOW wins (the safer choice).
The Driver Station shows "Drive speed" during the match.
Speeds are a first try: tune SLOW_MODE_SCALE, NORMAL_SCALE and TURBO_SCALE in DriverControls.java with the drivers.

Good to know

Directions match the robot: stick forward = drive forward, stick left = strafe left, right stick left = turn left.
Let go of the sticks and the robot coasts to a stop (FLOAT, decision 023).
Nothing moves during INIT (game rule G304.H).
Free buttons (A B X Y, dpad, stick presses) can get driver jobs later, e.g. "turret to forward".

Gamepad 2: OPERATOR

Runs the mechanisms. Pick it on the Driver Station with START + B. Write each job next to a button (pencil!), then tell the coders.

Left trigger (LT / L2)

Left bumper (LB / L1)

Dpad UP

Dpad LEFT

Dpad RIGHT

Dpad DOWN

Left stick

Left stick: press it down

Right trigger (RT / R2)

Right bumper (RB / R1)

Y (triangle)

X (square)

B (circle)

A (cross)

Right stick

Right stick: press it down

Jobs that need a button (for each: HOLD it, TOGGLE on/off, or PRESS once?)

[ ] Intake IN
[ ] Intake OUT
[ ] Water wheel UP
[ ] Water wheel DOWN (clear a jam)
[ ] Flywheel ON / OFF
[ ] FIRE (the trigger)
[ ] Flower flipper shake
[ ] Turret LEFT / RIGHT
[ ] Turret to FORWARD
[ ] Hood UP / DOWN
[ ] Turret AUTO-AIM (later, #7)

Tick each job once it has a button. A stick or trigger is good for "how much" (turret speed); a button is good for on/off.

Questions to decide

1. Intake: hold, or toggle on/off?
2. Fire: one button for the whole shot (load, fire, reset) or a button each?
3. Anything automatic? e.g. stop the intake when a NECTAR is detected
4. Lights: what should they show?